MODERN CRYPTOGRAPHY · MINI PROJECT

Lightweight Feistel-Based Smart Meter Encryption Suite

Problem analysis, feasibility, methodology and algorithm relevance

Smart-grid security · Symmetric-key cryptography · Web simulation

Academic Year: _____

Project and Team Details

Project

Lightweight Feistel-Based Smart Meter Encryption Suite

Course / Subject

Modern Cryptography
Programme: _____
Section: _____

TEAM MEMBERS

No.	Name	Enrollment No.	Primary Module
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____
4	_____	_____	_____
5	_____	_____	_____

Why Smart-Meter Telemetry Needs Protection

Residential meters send small, frequent usage reports to a utility gateway. These readings affect billing and can also reveal household activity patterns.



Security issue

If the link is observed or modified, an attacker may infer behaviour, alter readings or replay an old report.

Engineering issue

The meter has limited memory, processing capacity and energy. Protection must therefore be measurable as well as correct.

Problem Definition and Study Boundaries

Core problem

Design an understandable 8-bit Feistel cipher for a constrained smart-meter model, then measure how its cost and diffusion change with the number of rounds.

Q1

Correctness

Can every 8-bit reading be encrypted and recovered with the same key?

Q2

Diffusion

How close does the output come to changing four of eight bits after one input-bit flip?

Q3

Cost

What time and estimated energy are required by 4, 6 and 8 rounds?

Scope boundary

This is a teaching and evaluation model. An 8-bit block and a short key are not suitable for real billing security; production meters should use standard authenticated encryption.

Where the Proposed Cipher Fits

Existing approaches provide the baseline for the project; they are not treated as direct equivalents.

Approach	Strength	Limitation in this study
Unencrypted / checksum only	Very low processing cost	No confidentiality; checksum is not authentication
DES	Classic Feistel reference with 16 rounds	56-bit key is obsolete; 64-bit implementation is larger
AES-128	Modern standard; strong and widely supported	More complex than an 8-bit teaching model
Lightweight standard ciphers	Designed for constrained devices	Less transparent for a short classroom implementation
Proposed 8-bit Feistel	Small, traceable and easy to benchmark	Tiny block/key space; demonstration only

Reference baseline: DES structure (FIPS 46-3) and AES (FIPS 197).

Project Feasibility

Technical

Python supports direct bit operations. FastAPI exposes the simulation, while browser charts show internal states and results.

Computational

The complete input space has only 256 blocks, so exhaustive correctness tests and large avalanche samples are practical.

Hardware

The first version uses a documented energy model. C, Wokwi or Arduino measurement can be added without changing the cipher design.

Schedule

Modules are independent enough for five members to work in parallel, with shared test vectors used during integration.

Main risks | **Unfair timing comparison** | **Confusing estimates with measurements** | **Overstating security** | **Integration mismatch**

Control: identical test vectors, repeated timing runs, labelled estimates and explicit limitations.

Objectives and Expected Outputs

01

Build the cipher

Implement 8-bit encryption, decryption and a configurable 4/6/8-round engine.

02

Simulate meter data

Generate time-based kWh readings, quantize them and construct transmission packets.

03

Make it observable

Display plaintext, subkeys and left/right values for every round in the web UI.

04

Evaluate diffusion

Measure plaintext avalanche, key sensitivity and output-bit dependencies.

05

Measure cost

Report repeatable execution-time and estimated energy results for each round count.

06

Demonstrate controls

Show sequence validation, replay rejection and integrity checking around the cipher.

Methodology

1
Specify
Block, key, S-box, permutation and output convention

2
Implement
Reference cipher in Python with a round trace

3
Verify
Known-answer, boundary and exhaustive round-trip tests

4
Simulate
Meter readings, quantization and packet transmission

5
Evaluate
Timing, avalanche, key sensitivity and energy estimate

6
Present
FastAPI dashboard, charts and final comparison

Experimental controls

Use the same input sets for every round count; time only the cipher call; repeat runs; store raw results; report mean and variation; keep estimated and measured energy separate.

Success criteria

Correct recovery for all 256 plaintext blocks; reproducible benchmarks; avalanche results shown by round count; working end-to-end demonstration.

Why These Techniques Are Relevant

The design separates the experimental cipher from the controls needed by a communication system.

CORE CIPHER

Feistel network

Provides reversible encryption while allowing a simple custom round function.

S-box substitution

Introduces non-linearity so output changes are not a direct linear mapping of the input.

Bit permutation

Moves S-box output bits to improve diffusion across later rounds.

Rotating key schedule

Derives a different 4-bit subkey for each round with low implementation cost.

SUPPORTING CONTROLS

Nonce / IV

Reduces visible repetition when similar readings are encrypted.

Sequence number

Allows the gateway to identify stale or replayed reports.

Integrity tag

Detects packet modification; it remains separate from encryption.

Encryption provides confidentiality; it does not provide integrity or replay protection by itself.

Link Between Design Choices and Evaluation

Design choice	Reason for inclusion	How it is evaluated
8-bit block	Matches a quantized sample and keeps internals visible	Exhaustive input tests; repetition demo
4 / 6 / 8 rounds	Creates a clear cost-versus-diffusion variable	Timing, energy estimate and avalanche score
4-bit S-box	Adds non-linearity with a small lookup table	Input/output difference heatmap
Permutation	Spreads changed bits into later rounds	Per-round output-bit sensitivity
16-bit baseline key	Simple schedule for the experiment	Key-flip sensitivity and brute-force limitation

Expected conclusion

More rounds should improve diffusion but increase processing cost. The value of the project is the measured trade-off—not a claim that the custom cipher is production-ready.